TENGTENG LEI

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RESEARCH INTERESTS

Piezoelectric tactile sensor array, e-skin, smart sensor, metal-oxide thin-film transistors (MO TFTs), thin-film circuit design, spiking neural network (SNN), in-memory computing, transformer accelerator, large language models (LLMs)

EDUCATION

Ph.D. at The Hong Kong University of Science and Technology (HKUST) *08/2020-12/2023*
Department of Electronic and Computer Engineering
Supervisor: Prof. Man Wong

M. Eng. at Peking University (PKU) *08/2017-06/2020*
School of Electronic and Computer Engineering
Supervisor: Prof. Shengdong Zhang

B. Eng. at Sun Yat-sen University *08/2013-06/2017*
School of Opto-electronics and Information Technology

WORK EXPERIENCES

Research Associate at King's College London & Northeastern University London *02/2025-Now*
Line Manager: Prof. Bipin Rajendran

Research Associate at The Hong Kong University of Science and Technology *01/2024-01/2025*
Supervisor: Prof. Man Wong

Intern at HUAWEI Technologies Co., Ltd. *06/2019-08/2019*

RESEARCH EXPERIENCES

■ **Hardware-Algorithm Co-Design for Next Generation AI | KCL**
Brain-Inspired Learning Accelerators for Spiking LLMs
- Project Goal: Tape-out a SRAM-based in-memory computing chip to support on-chip learning using the memory-efficient zeroth-order (MeZO) optimizer
- Contribution: Designed a hardware-efficient event-triggered perturbation generation unit (PGU) to support the MeZO optimizer, enabling on-chip learning for a spiking transformer accelerator (tape-out planned)

■ **Smart Sensor System (Device → Circuit → System) | HKUST**
Tactile Sensor Arrays and Artificial Neural Networks (ANNs)
- Pioneered the modeling and fabrication of dual-gate metal-oxide TFTs, unlocking new degrees of freedom in analog circuit design
- Developed a high-performance active-matrix tactile sensor array featuring in-pixel amplification and non-uniformity compensation circuits
- Demonstrated monolithic integration of tactile sensor arrays with hardware ANNs for edge computing and in-sensor computing
- Built monolithically integrated transimpedance current accumulators for analog IMC based on DG TFTs

■ **Thin-Film Circuit Design | PKU**
Gate Driver on Array (GOA) for Flat Panel Display (FPD)
- Invented a highly reliable GOA design robust against significant threshold voltage shifts in metal-oxide TFTs
- Conceived and demonstrated a GOA with a dynamic inverter to generate multiple scan/emission signals for AMOLED displays with internal compensation.
- Designed a high-speed GOA with a fully bootstrapped circuit topology for large size 8K LCDs

■ **IC Tape-Out Experiences**
- TSMC 16nm FinFET | Spiking Transformer Accelerator | KCL
- SMIC 0.18 µm CMOS | Driver IC for Micro-OLED Display | Gamma Correction DAC | PKU

RESEARCH PROJECTS

[1] **ICDT Huada Jiutian Cup Innovation Competition Project**: Full-Flow Design of Gate Driver on Array (GOA) Circuit for Flat-Panel Displays, 2022, ¥40,000 Lead

[2] **Advanced Research and Invention Agency (ARIA, UK)**: Brain-Inspired Learning Accelerator for Large Language Models (LLMs), 2024 - 2027, £3M Participant

[3] **Innovation and Technology Fund Project**: Convolution Neural Network Architecture Based on Metal-Oxide Thin Film Transistor Technology for Pattern Recognition, 2023 - 2025, HK$1.12M Participant

[4] **Innovation and Technology Fund Project**: Research on In-Display Fingerprint Sensing Technology Based on Oxide Thin-Film Transistors (TFTs), 2023 - 2025, HK$1.07M Participant

PUBLICATIONS

Journal Papers

[1] **Tengteng Lei**, Prabodh Katti, Rashi Dutt, Houssem Sifaou, Tan Peng, Osvaldo Simeone, Kai Xu, Bipin Rajendran, "Event-triggered Implicit Perturbation for Zeroth-Order Optimization of Spiking Transformers," *IEEE Transactions on Circuits and Systems-I*, under review.

[2] **Tengteng Lei**, Boyi Zhu, Yushen Hu, Tianming Li, and Man Wong, "A Flexible Dual-Gate TFT-Based Tactile Sensor Array with Polarity-Aware Encoding for Neuromorphic Recognition," *FlexMat*, major revision.

[3] **Tengteng Lei**, Jiayi Mao, Yushen Hu, and Man Wong, "Monolithically integrated transimpedance accumulators for Analog In-Memory Computing based on dual-gate metal-oxide thin-film transistors," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 16, no. 2, pp. 607-616, 2026.

[4] Xinying Xie, Qining Leo Wang, Runxiao Shi, **Tengteng Lei**, Chang-Jin "CJ" Kim and Man Wong, "An active-matrix digital microfluidic device based on surfactant-mediated electro-dewetting," *Lab Chip*, 2026, doi: 10.1039/d5lc00992h. (Corresponding author)

[5] **Tengteng Lei**, Yushen Hu, Xinying Xie, Runxiao Shi, and Man Wong, "In-sensor computing-based smart sensing architecture implemented using a dual-gate metal-oxide thin-film transistor technology," *Adv. Electron. Mater.*, vol. 11, no. 9, 2025, 2400572.

[6] Yushen Hu, Xinying Xie, **Tengteng Lei**, Runxiao Shi, Man Wong, "A sliding-kernel computation-in-memory architecture for convolutional neural network," *Adv. Sci.*, vol. 11, no. 46, 2024, 2407440. (Corresponding author)

[7] **Tengteng Lei**, Runxiao Shi, Zong Liu, Yushen Hu, and Man Wong, "Neuromorphic sensor-perception systems for immersive displays," *IEEE open journal on immersive displays*, vol. 1, pp. 20-27, 2024.

[8] Runxiao Shi, **Tengteng Lei**, Zhihe Xia, and Man Wong, "Low-temperature metal-oxide thin-film transistor technologies for implementing flexible electronic circuits and systems," *Journal of Semiconductors*, 44, 091601, 2023. (Co-first author)

[9] **Tengteng Lei**, Runxiao Shi, Yuqi Wang, Zhihe Xia, and Man Wong, "A Comparative study on inverters built with dual-gate thin-film transistors based on intrinsically depletion- or enhancement-mode technologies," *IEEE Trans. Electron Devices*, vol. 69, no. 8, pp. 3186-3191, 2022.

Conference Papers

[1] **Tengteng Lei**, Boyi Zhu, Yushen Hu, and Man Wong, "Accurate acquisition of pressure signals with optimal amplitude using flexible tactile sensor array," *in Digest Tech. Papers Transducers'25*, Orlando, June 29-July 3, pp. 1621-1624, 2025. [Flagship Conference in Sensors & Microsystems]

[2] **Tengteng Lei**, Yushen Hu, Xinying Xie, and Man Wong, "An active-matrix piezoelectric tactile sensor array with in-pixel amplifier and non-uniformity compensation," *in Digest Tech. Papers Transducers'23*, Kyoto, June 25-29, pp. 298-301, 2023. [Oral Paper] [Flagship Conference in Sensors & Microsystems]

[3] **Tengteng Lei**, Yushen Hu, and Man Wong, "A tactile sensor array with a monolithically integrated neural network for edge computation," *in Digest Tech. Papers MEMS'23*, Munich, Jan 15-19, pp. 13-16, 2023. [Oral Paper] [Flagship Conference in Sensors & Microsystems]

[4] **Tengteng Lei**, Yushen Hu, and Man Wong, "Active-matrix tactile sensor array based on the monolithic integration of PVDF and dual-gate transistors," *in Digest Tech. Papers MEMS'22*, Tokyo, Jan 09-13, pp. 71-74, 2022. [Oral Paper] [Flagship Conference in Sensors & Microsystems]

[5] **Tengteng Lei**, Congwei Liao, Jie Huang, Shengdong Zhang, "A robust a-IGZO TFT integrated scan/emission driver with dynamic inverter for AMOLED display," *SID Symposium Digest of Technical Papers*, 51 (1): 1354-1357, 2020.

[6] **Tengteng Lei**, Congwei Liao, Jie Huang, Jiwen Yang, Shengdong Zhang, "Oxide thin film transistors integrated DC-DC converter with high efficiency for passive RFID tag," *SID Symposium Digest of Technical Papers*, 50 (1): 1660-1663, 2019.

[7] **Tengteng Lei**, Congwei Liao, Jie Huang, Ying Wang, Shengdong Zhang, "An a-InGaZnO TFT gate driver circuit with positive feedback effect," *International Conference on CAD-TFT*, 2018. [Oral Paper]

Patents

[1] **Tengteng Lei**, Man Wong. A hysteresis comparator implemented using dual-gate thin film transistors. CN120512123A, Chinese Patent.

[2] Xuchi Liu, **Tengteng Lei**, Man Wong. A dynamic offset-cancellation circuit based on dual-gate thin-film transistors. CN116800209A, Chinese Patent.

[3] Shengdong Zhang, **Tengteng Lei**, Congwei Liao, Jie Huang. A high-speed gate driving unit and a circuit. CN109859669B, Chinese Patent. Granted: September 02, 2022.

[4] Shengdong Zhang, **Tengteng Lei**, Congwei Liao, Jie Huang. Voltage converter and radio frequency identification device. CN109492740B, Chinese Patent. Granted: March 01, 2022.

[5] Shengdong Zhang, **Tengteng Lei**, Congwei Liao, Jie Huang. Shift register unit, gate drive circuit and display device. CN109285505B, Chinese Patent. Granted: June 23, 2020.

HONORS & AWARDS

2023	PG Workshop Best Oral Award
	HKUST RedBird Academic Excellence Award
	Senior Teaching Assistant Fellowship
2022	Grand Prize in the "ICDT Empyrean Cup Innovation Competition"
	Solomon Systech Scholarship
2020	Excellent Graduate at Peking University
2019	Founder Scholarship
	Merit Student at Peking University
2016	87th Physics Scholarship
	First Honors Scholarship
	Honorable Mention for Mathematical Contest in Modeling
	Duke Kunshan University Medical Physics Young Talent
2015	First Honors Scholarship
	Outstanding Member of Sun Yat-sen University
2014	Second Honors Scholarship

TECHNICAL SKILLS

IC Design:	Silvaco Gateway, Cadence Virtuoso/Xcelium/Genus/Innovus/Pegasus/Tempus/Joules
Programming:	Python, PyTorch, Verilog HDL, LabVIEW
Embedded System:	Vivado, FPGA, STM32, PYNQ-Z2
Micro-Fabrication:	Photolithography, Dry/Wet Etching, Sputtering/PECVD Deposition
Language:	Mandarin (Native), English (Fluent)